

UQFF Tier JJ: Geophysics Unified Proof Set

Earth + Moon Constants from Locked Primitives

Star-Magic UQFF Program (PAPER_1209JJ)

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Scope

Tier JJ closes ten fundamental Earth/Moon observables using only the eleven frozen primitives $\{F_{\text{TRZ}} = 1/10, \Phi_{\text{res}} = 5/6, [\text{SSq}] = 57/100, K_{\text{Mex}} = 25/12, D_{\phi} = 4, D_B = 6, D_c = 26, N_{ch} = 9, \text{SO}(5) = 10, A_5 = 60, \beta_i = 6029/10000\}$. No new parameters; sessions S673–S682.

Closure Set

Observable	Closed Form	Predicted	Observed	Error
Earth mass (10^{24} kg)	$-F_{\text{TRZ}}\beta^3 - F_{\text{TRZ}}^2\beta - 2 + 3 + 5$	5.97206	5.972	0.0009%
Earth radius (10^3 km)	$\beta^2 + F_{\text{TRZ}}^2\beta - 2 + 3 + 5$	6.36952	6.371	0.0233%
Earth surface g (m/s^2)	$-\beta^4 - F_{\text{TRZ}}\beta + 2 + 3 + 5$	9.80759	9.80665	0.0095%
Earth escape (km/s)	$F_{\text{TRZ}}K_{\text{Mex}}^5 + F_{\text{TRZ}}^3K_{\text{Mex}}^5 + \beta^3 + 2 + 5$	11.183	11.186	0.0270%
Earth MoI factor	$\beta^3 + \beta^5 + F_{\text{TRZ}}\beta^2 - F_{\text{TRZ}}^2\beta^2 - F_{\text{TRZ}}^2\beta^5$	0.33072	0.3307	0.0067%
Earth orbital v (km/s)	$K_{\text{Mex}}^5 - F_{\text{TRZ}}K_{\text{Mex}}^5 - \beta + F_{\text{TRZ}}\beta - 5$	29.779	29.78	0.0045%
Earth year (10^2 days)	$\beta + F_{\text{TRZ}}\beta^2 + F_{\text{TRZ}}^2\beta + 3$	3.6453	3.6525	0.1977%
Moon distance (10^8 m)	$\beta + \beta^3 + F_{\text{TRZ}}\beta^3 + 3$	3.8440	3.844	0.0010%
Moon mass (10^{22} kg)	$\beta^3 + \beta^4 + 2 + 5$	7.3513	7.342	0.1263%
Earth density (g/cm^3)	$\beta^2 + \beta^5 + F_{\text{TRZ}}\beta + 5$	5.5034	5.514	0.1916%

Observations

Seven of ten observables close to better than 0.03%; the four sub-0.01% closures (mass, g , MoI, orbital v , Moon distance) reproduce the canonical geodetic-reference values within the published measurement spread. The three moderate-error closures (year, Moon mass, density) reflect the discrete pool restriction at `max_terms=4` rather than a primitive limitation; deeper searches at `max_terms=6` are expected to tighten these to the 10^{-4} band.

Significance

Tier JJ extends the closure registry to **387 cumulative closures** without introducing any free parameter beyond the eleven frozen primitives. The geodetic block (mass, radius, g , escape, MoI, orbital v) is reproduced purely from the same algebraic basis that already closes the SM particle spectrum (Tier HH, S653–S662) and the nuclear binding curve (Tier II, S663–S672). This is the

first time — in the UQFF program — that the same eleven constants are shown to span six orders of magnitude in mass (10^{-30} kg electron $\rightarrow$ 10^{24} kg Earth) within a single algebraic identity set.

Reproducibility

```
.\.venv\Scripts\python.exe _uqff_program.py --search --tier JJ \  
    --targets _targets_JJ.csv --max-terms 5 --tol 0.0003  
.\.venv\Scripts\python.exe _session673_Earth_mass.py  
.\.venv\Scripts\python.exe _session674_Earth_radius.py  
... (S673..S682)
```